Asaad Hanna

Quantum Algorithms | Many-Body Simulation | Fundamental Physics

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Research Profile

Physics MSc student working on quantum algorithms, with experience in fermionic many-body simulation and classical numerical benchmarks. Research interests center on quantum simulation of lattice gauge theories and applications to nuclear and particle physics, building on current work in operator functions, Hamiltonian simulation, and Hubbard-model Green's functions.

Education

MSc in Physics
Concordia University, Montreal — Sep. 2025 – Aug. 2027 (expected)
Research in quantum algorithms; supervisor: Dr. Abhijeet Alase.

BSc, Honours in Physics, with Distinction
Concordia University, Montreal — 2022 – 2025
Minor in Computer Science.

Research Experience

Quantum Software Developer Intern
AlgoLabs, Université de Sherbrooke | Dr. Simon Verret — May – Aug. 2026

- Extended a Liouvillian-recursion Green's-function workflow to three- and five-site Hubbard models with degenerate ground states.
- Implemented fermionic-swap circuits connecting spin-related ground states; combined pure-state contributions to evaluate ensemble observables without directly preparing a mixed state.
- Compared Green's functions and Galitskii–Migdal energy estimates with exact-diagonalization benchmarks, using Qiskit simulators and IBM quantum hardware.
- Investigated symmetry-based qubit tapering and implemented a symplectic-method demonstration to explore qubit reduction.

Research code: Liouville-Lanczos, asaad branch.

MSc Research — Quantum Algorithms for Operator Functions
Concordia University | Dr. Abhijeet Alase — 2025 – present

- Investigating accuracy and resource costs of algorithms that implement functions of unitary operators, a primitive used across quantum applications.
- Comparing Laurent approximation and circuit constructions through mathematical analysis and computational benchmarks, distinguishing approximation error, query complexity, and implementation overhead.
- Formulating bounds and optimality questions for different function classes and applications; these are ongoing research directions.

Honours Thesis — Statistical Mechanics and Machine Learning
Concordia University | Dr. Saurabh Maiti — 2024 – 2025

- Built numerical demonstrations of equilibration and entropy in statistical ensembles; explored connections between statistical mechanics and Boltzmann machines.
- Developed simulations and visualizations to explain equilibrium distributions and learning dynamics.

Additional Research Experience

Optical Microresonators
Concordia University | Dr. Pablo Bianucci 2023 – 2024

- Performed fabrication, experiments, measurements, and data analysis for a systematic study of flame-fabricated surface nanoscale axial photonic resonators; first author of the resulting 2024 conference publication.

Research courses: Twisted bilayer graphene fabrication with Dr. Alexandre Champagne (2025); symmetries of Hermitian matrices with Dr. Saurabh Maiti (2024), Concordia University.

Publications

Published conference paper
A. Hanna, E. C. V. Eadie, S. Deep, and P. Bianucci. Systematic study of Flame-Fabricated Surface Nanoscale Axial Photonics resonators. *Advanced Photonics Congress 2024*, paper SoTh3D.3 (Optica, 2024). doi:10.1364/SOF.2024.SoTh3D.3.

Journal manuscript under review
S. Padvaiskas, E. Eadie, S. Sherr, **A. Hanna**, S. Deep, and P. Bianucci. Systematic Study of Flame-heating Fabrication of Surface Nanoscale Axial Photonic Resonators. *Advanced Optical Technologies*, submitted September 2026; under review.

Teaching Experience

Teaching Assistant — Physics
Concordia University Jan. 2025 – present

- Held successive appointments each semester: Electricity and Magnetism grading for approximately 150 students; quiz writing for Energy and Environment; advanced laboratory instruction and report assessment.
- Currently lead Electricity and Magnetism tutorials, guiding students through calculations and physical reasoning.

Awards & Research Funding

Hydro-Québec Graduate Award (2025–2026); **NSERC** Undergraduate Summer Research Award (2025); **Concordia** undergraduate research awards (2023, 2024).

Fields Institute visitor grant awarded for the 2026 thematic program on quantum algorithms for differential equations; not taken up.

Technical Skills & Training

Scientific computing: Python, NumPy, Qiskit, Qiskit Nature, PennyLane; exact diagonalization, fermion-to-qubit mappings, and numerical benchmarking.

Additional tools: Mathematica, MATLAB, Git, Linux, Bash, LaTeX, and Manim.

Training: AlgoLabs Summer Course on Quantum Algorithms, Université de Sherbrooke (2025).

Languages: English, French, and Arabic.

Leadership & Scientific Communication

Founder and Lead, Quantum Computing Club (2025–2026): delivered quantum-algorithm workshops, prepared undergraduates for hackathons, and organized researcher interviews and outreach.

Presented *The Equilibrating Behaviour of Statistical Systems* at INTRIQ and the Concordia Undergraduate Research Showcase (2025); additional poster presentation at AQC UdeS.